

Marine Oil Pollution Monitoring Based on a Morphological Attention U-Net Using SAR Images

Lena Chang Yi-Ting Chen Ching-Min Cheng
Yang-Lang Chang Shang-Chih Ma

Sensors, 2024

Abstract

This paper proposes an improved oil spill detection framework based on a Morphological Attention U-Net using synthetic aperture radar (SAR) images. The method enhances the full-scale aggregated MobileUNet (FA-MobileUNet) by introducing a morphological attention module (MAM) and label smoothing during training. The proposed approach effectively reduces fragmented regions and holes in oil spill segmentation results, producing more complete and reliable oil pollution maps. Experimental results on real oil spill incidents demonstrate significant performance improvements over conventional U-Net-based models.

1 Introduction

Marine oil pollution is a major environmental concern caused by offshore platforms, ship accidents, illegal discharges, and pipeline leaks. Traditional monitoring using ships or aircraft is limited by coverage, weather, and illumination conditions.

Synthetic aperture radar (SAR) provides all-weather, day-and-night monitoring capability. Oil spills suppress capillary waves on the sea surface, resulting in reduced radar backscatter and dark regions in SAR images. However, look-alike phenomena such as low-wind areas and biogenic slicks complicate oil spill detection.

Deep learning methods, especially U-Net-based semantic segmentation models, have shown strong performance in SAR oil spill detection. Despite this progress, segmen-

tation results often suffer from fragmented regions and incomplete oil spill shapes due to non-uniform sea clutter and speckle noise.

This study addresses these issues by introducing morphological concepts into the attention mechanism of FA-MobileUNet.

2 Related Work

Early oil spill detection methods relied on thresholding, statistical tests, and handcrafted features. Polarimetric and texture-based approaches improved discrimination but remained sensitive to environmental variability.

Recent deep learning approaches replace handcrafted features with automatically learned representations. U-Net and its variants dominate SAR oil spill segmentation due to their encoder–decoder structure and skip connections.

FA-MobileUNet improved oil spill detection by integrating CBAM attention, atrous spatial pyramid pooling (ASPP), and full-scale aggregation. However, fragmentation and holes in detection results remain a challenge, motivating further architectural improvements.

3 Oil Spill Dataset

The study uses the extended MKLab oil spill dataset derived from Sentinel-1 SAR imagery. The dataset contains 1239 images, including 1129 training images and 110 testing images.

Images are VV-polarized C-band SAR with a spatial resolution of 10 m. Each image includes five semantic categories: oil spills, look-alikes, ships, sea surface, and land. All images are resized to 352×352 pixels for model training.

4 FA-MobileUNet Architecture

FA-MobileUNet follows an encoder–decoder architecture derived from U-Net. The encoder backbone is replaced with MobileNetV3 to reduce computational complexity while preserving feature extraction capability.

The model integrates three key components:

- Convolutional Block Attention Module (CBAM)
- Atrous Spatial Pyramid Pooling (ASPP)
- Full-scale Aggregation (FA) module

These components enable effective multiscale feature extraction and improve discrimination between oil spills and look-alikes.

5 Morphological Attention Module

Standard CBAM uses convolution-based spatial attention, which may fail to preserve shape completeness in SAR images. To address this limitation, the spatial attention module is replaced with a Morphological Attention Module (MAM).

Morphological operations such as dilation, erosion, and closing emphasize structural shape information. The closing operation smooths boundaries and fills small holes inside oil spill regions.

Given an input feature map F , the morphological attention output is expressed as:

$$MM(F) = F \times W_1 \times K_c$$

where W_1 is a 1×1 convolution and K_c denotes the morphological closing operation.

The modified CBAM applies MAM to shallow encoder layers to enhance spatial completeness while preserving semantic detail in deeper layers.

6 Loss Function and Label Smoothing

Categorical cross-entropy loss is used for multi-class segmentation. Due to severe class imbalance, label smoothing is applied to reduce overconfidence toward majority classes.

The smoothed label is defined as:

$$y_{ls} = (1 - \alpha)y + \frac{\alpha}{K}$$

where α is the smoothing factor and K is the number of classes.

Label smoothing improves generalization and enhances detection performance for minority categories such as oil spills and look-alikes.

7 Methodology

The complete oil pollution monitoring methodology consists of the following steps:

1. Preprocess Sentinel-1 SAR images and resize inputs.
2. Extract multiscale features using MobileNetV3 encoder.
3. Apply ASPP to capture multiscale contextual information.
4. Aggregate features using full-scale aggregation.
5. Enhance spatial attention using morphological attention modules.
6. Train the network with categorical cross-entropy and label smoothing.
7. Generate pixel-level oil spill segmentation maps.

This integrated framework improves spatial completeness and robustness under sea clutter.

8 Experimental Results

The model is trained for 1000 epochs using the Adam optimizer. Experiments are conducted on a Windows-based system with GPU acceleration.

Training convergence shows high accuracy and low loss, indicating stable optimization. Visual results demonstrate that the improved FA-MobileUNet produces smoother oil spill regions with fewer holes and fragmented areas compared to baseline models.

9 Performance Analysis

Quantitative evaluation uses Intersection over Union (IoU) and mean IoU (mIoU). The original FA-MobileUNet achieves an mIoU of 83.74%.

By applying morphological attention to shallow encoder layers and using two iterations of morphological closing, the mIoU improves to 83.94%. With label smoothing, the final model achieves an mIoU of 84.55%, representing a 17.15% improvement over the original U-Net.

Comparisons with U-Net, LinkNet, PSPNet, DeepLabv2, DeepLabv3+, and other recent models confirm that the proposed method achieves the best overall segmentation performance.

10 Conclusion

This paper presents a Morphological Attention U-Net for marine oil pollution monitoring using SAR images. By integrating morphological operations into the attention mechanism and applying label smoothing, the proposed model significantly improves oil spill segmentation completeness and accuracy.

Experimental results validate the effectiveness of the approach in real-world oil pollution incidents. The framework provides a robust solution for operational marine oil spill monitoring and can be extended to other SAR-based environmental applications.